

David Berghaus

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Senior Research Scientist in machine learning, with a background in physics, math and numerical algorithms. Research on foundation models for scientific inference (published at ICML, NeurIPS, and ICLR) and on LLM reasoning such as looped transformers and latent chain-of-thought, alongside applied industry work building LLM-based systems.

Experience

Senior Research Scientist

2025 – Present

Fraunhofer IAIS

Sankt Augustin, Germany

- Develop foundation inference models for scientific dynamics (ODEs, SDEs, and temporal point processes), with publications at ICML, NeurIPS, and ICLR.
- Work on LLM reasoning, including looped transformers and latent (continuous) chain-of-thought, with several works presented at the ICLR Latent & Implicit Thinking workshop.

Research Scientist

2023 – 2025

Fraunhofer IAIS

Sankt Augustin, Germany

- Developed foundation inference models for Markov jump processes (NeurIPS 2024), the basis for the later work above.
- Fine-tuned a large language model for German legal tasks (compliance checks, document analysis and summarization).
- Built LLM-based systems for industry clients, including automated email/invoice order processing and an LLM recommender for patient matching in personalized medicine.
- Led client-acquisition talks and consulted on AI-driven workflow optimization.

IT Research Assistant

2019 – 2023

Physikalisches Institut, University of Bonn

Bonn, Germany

- Designed and developed room-booking and event-management software on Indico.
- Automated installation and maintenance of Windows and macOS machines; evaluated and documented internal software.

Co-Founder

2016 – 2019

Ems und Berghaus App-Entwicklung GbR

Bonn, Germany

- Co-founded a startup building a social app for discovering nearby events; led the project and a team of freelance designers and developers, and shipped the first Android version.

Education

Ph.D. in Mathematical Physics / Computational Number Theory

2021 – 2023

University of Bonn

Bonn, Germany

Thesis: *On the Numerical Computation of Modular Forms* ([link](#)), *magna cum laude*. Developed numerical algorithms (Python, Cython, C) and [databases](#) for special mathematical objects.

M.Sc. in Physics

2018 – 2020

University of Bonn

Bonn, Germany

Thesis on computing Laplacian eigenvalues at arbitrary-precision arithmetic. Awarded the Bonn-Cologne Graduate School (BCGS) Honors Branch scholarship for excellent academic achievement.

B.Sc. in Physics

2015 – 2018

University of Bonn

Bonn, Germany

Thesis on Laplacian eigenvalues of regular polygons and multiple zeta values.

Selected Publications

90+ citations, h-index 6 (Google Scholar).

- *Foundation Inference Models for Ordinary Differential Equations*, [ICML](#) 2026.
- *In-Context Learning of Temporal Point Processes with Foundation Inference Models*, [ICLR](#) 2026. [\[first author\]](#)
- *Is Continuous CoT Better Suited for Multilingual Reasoning?*, [ICLR](#) 2026 (Latent & Implicit Thinking). [\[senior author\]](#)
- *Foundation Inference Models for Stochastic Differential Equations*, [NeurIPS](#) 2025.
- *Vision vs. Text-Based Parsing: Benchmarking LLM Strategies for Invoice Processing*, [IEEE Big Data](#) 2025. [\[first author\]](#)
- *Fine-Tuning Large Language Models for Compliance Checks*, [IEEE Big Data](#) 2024. [\[senior author\]](#)
- *Foundation Inference Models for Markov Jump Processes*, [NeurIPS](#) 2024. [\[first author\]](#)
- *On the Computation of Modular Forms on Noncongruence Subgroups*, [Mathematics of Computation](#) 2022. [\[first author\]](#)

Skills

Languages: Python (advanced), Julia (advanced), Cython (advanced), C/C++.

ML / Tools: PyTorch, TensorFlow, HuggingFace, LLM fine-tuning, Git/GitHub, Linux, L^AT_EX.

Spoken: German (native), English (fluent).